

Value creation and value appropriation using voice AI technology: integrating B2B and B2C marketplace

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Abstract

Purpose – While Voice AI technology (VAI) adoption has accelerated, firms struggle to create and appropriate value. This study aims to address the critical gap between widespread VAI adoption and elusive monetization by developing an integrated framework that guides strategic implementation decisions. It examines how firms should integrate B2B and B2C marketplaces to create value with VAI and identify optimal approaches for value appropriation through build versus buy decisions.

Design/methodology/approach – This study employed a rigorous grounded theory methodology with methodological triangulation across three studies: consumer surveys (Study 1; $n = 197$), 18 structured focus groups with business professionals (Study 2) and in-depth expert interviews with decision-makers responsible for VAI implementation (Study 3). This systematic approach ensured comprehensive theory development validated through multiple stakeholder perspectives.

Findings – The framework used in this study reveals several counterintuitive relationships challenging conventional technology implementation. The authors found that higher integration potential paradoxically leads to lower investment in proprietary solutions as firms leverage existing platforms. Regulatory constraints emerge as powerful counterforces that override otherwise compelling business incentives. Domain specialization significantly enhances perceived value and competitive advantage. Anthropomorphism produces different effects across use contexts in B2C market, requiring calibrated design approaches rather than uniform implementation.

Research limitations/implications – Longitudinal research examining VAI's influence on business value, customer acquisition and retention through behavioral data analysis would significantly advance understanding of long-term value creation mechanisms. Similarly, cross-cultural studies comparing VAI adoption patterns across regions with different privacy orientations, technological infrastructures and cultural attitudes toward AI could reveal important contextual boundaries to our framework. Finally, sector-specific investigations testing our framework across manufacturing, financial services, healthcare and hospitality would enhance its practical utility while potentially revealing industry-specific modifications.

Practical implications – The framework used in this study provides actionable guidance across strategic positioning of VAI capabilities, implementation approach including investment sequencing and context-



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sensitive consumer experience design. These insights enable organizations to make more informed decisions, ultimately enhancing value creation and appropriation.

Social implications – The findings highlight important trade-offs between regulatory protections and business innovation, offering policymakers insights for developing balanced frameworks that safeguard consumer interests while enabling technological advancement.

Originality/value – The paper proposes an integrated framework for VAI value creation and monetization. It demonstrates bidirectional influence of B2B and B2C marketplace for implementation and monetization success. This study makes three significant contributions: bridging information systems and marketing perspectives, extending technology adoption research beyond initial acceptance to examine counterintuitive factors influencing postadoption value extraction and revealing how the interplay between B2B and B2C marketplaces creates unique dynamics in VAI implementation.

Keywords Voice AI technology, Artificial intelligence, Value creation, Value appropriation, Interaction orientation, Business models, Regulatory environment, Anthropomorphism, Technology adoption

Paper type Research paper

1. Introduction

"Hey Alexa, what's the weather today?" For millions of consumers globally, this simple interaction represents a fundamental shift in how humans engage with technology. In boardrooms across sectors, executives are urgently asking: How do we transform these everyday voice interactions into sustainable revenue and customer value?

For technology transformations, 2023 rivals watershed moments like the emergence of internet browsers or the iPhone launch, with AI mentions peaking in corporate earnings – reflecting both managerial preoccupation and market relevance. Within this AI revolution, Voice AI (VAI) represents a disruptive frontier (Kaplan, 2018), enabling natural language interactions that eliminate traditional barriers to technology adoption, such as literacy, technical proficiency and physical limitations.

Voice AI (VAI) is the broad body of tools and technologies that use AI to drive synthetic voice interfaces. Amazon's Alexa, Google's Hey Google or Apple's Siri are a few examples of VAI. They embody AI's ability to understand human thought processes or sequences and mimic human speech to participate in such processes (Li and Suh, 2021). They are sophisticated systems that learn from experiences, adapt responses to the decision or task at hand and blend frictionlessly with human decision-making (McCarthy, 2007). Such tools are almost always:

- responsive and, when necessary, conversational;
- superhuman-like in terms of speed, accuracy, knowledge and memory;
- empathic; and
- reliable learner.

They can often analyze vast amounts of data almost instantly, surpassing the typical human cognitive capacity.

This transformation presents a striking paradox: the VAI market shows promising growth, valued at \$12bn in 2022 with projections reaching \$50bn by 2029 (Fortune Business Insights, 2023), expanding at a commanding 21.8% CAGR between 2022 and 2032 (Market.us, 2023). Budzinski *et al.* (2019) suggested that voice assistants can significantly reduce transaction costs and the cost of information search for firms, and it is estimated that most businesses will gain from voice assistants, with a 20% reduction in customer service costs (Capgemini, 2019). Yet profitability remains frustratingly elusive as Microsoft discontinued Cortana, and Amazon's Alexa reported devastating losses of \$10bn in 2022 (The Economic Times, 2023, 2022). This

disconnect between widespread adoption and sustainable monetization reveals a critical strategic challenge facing organizations implementing voice technology.

As Voice AI gains importance, it is currently experiencing a chasm between early adopters and the early majority (Shin *et al.*, 2018), which stresses the need to understand the factors constraining the adoption and appropriation of Voice AI (Rabassa *et al.*, 2022). The difficulty stems from a fundamental knowledge gap – despite strong business interest and growing consumer adoption, firms lack established frameworks for VAI value creation and appropriation. This absence of strategic guidance arises from three distinct factors: limited widespread AI adoption, AI's expected domain-agnostic relevance, and concurrent experimentation across industries with minimal interorganizational learning. Extant research has concentrated primarily on initial VAI adoption factors – examining what drives consumers to first try voice assistants and what influences firms to experiment with voice technology (Moriuchi, 2019; McLean and Osei-Frimpong, 2019). However, these studies typically conclude their investigation at the point of initial acceptance or trial implementation, leaving critical questions unanswered: How do organizations sustain value creation after initial adoption? What mechanisms enable firms to monetize VAI investments beyond basic cost savings? How do ongoing user engagement patterns influence long-term business value?

This postadoption gap becomes particularly pronounced when examining B2B versus B2C value appropriation mechanisms. While consumer-focused research identifies engagement drivers like anthropomorphism and trust (Rhee and Choi, 2020), it fails to examine how these consumer behaviors translate into value for B2B marketplace. Conversely, business-focused studies explore operational efficiencies and customer service improvements but ignore how end-user adoption patterns constrain or enable organizational monetization strategies. Previous research stops short of examining these complex dynamics for three reasons: studies focus on single contexts (mostly B2C) rather than their interdependencies in B2B and B2C marketplaces. Investigations conclude at adoption rather than extending to value appropriation, especially in B2B markets, and existing frameworks cannot accommodate the counterintuitive relationships we observe in VAI implementation.

Our research addresses this gap by examining two essential questions that organizations must answer to strategically implement VAI:

- Q1. How should firms integrate VAI across B2B and B2C marketplaces to create sustainable value beyond initial adoption?
- Q2. How can firms appropriate this value through optimal build versus buy approaches that account for the complex interdependencies between business and consumer contexts?

Through a rigorous grounded theory methodology combining consumer surveys, managerial focus groups and in-depth expert interviews, we develop a comprehensive framework that challenges conventional wisdom about VAI implementation. Our findings reveal counterintuitive relationships between integration potential and investment levels, demonstrate how regulatory constraints create forces that often override traditional business incentives, and establish how domain specialization significantly enhances value creation potential. Most surprisingly, we discover that anthropomorphism – attributing human characteristics to technology – produces dramatically different effects across use contexts, challenging the common practice of treating VAI as a singular technology with uniform adoption drivers.

We bridge the gap between information systems and marketing perspectives by developing an integrated framework that explains value creation and appropriation

mechanisms in VAI implementation. This research makes three significant contributions to existing literature. First, we extend technology adoption research beyond initial acceptance to reveal the counterintuitive postadoption value extraction mechanisms that determine long-term VAI success – discovering that higher integration potential paradoxically leads to lower investment intensity. Second, we bridge the gap between B2B and B2C perspectives by demonstrating how consumer engagement patterns create “pull” effects that fundamentally reshape organizational investment strategies in ways that neither business nor consumer research has previously identified. Third, we reveal how anthropomorphism produces opposite effects across use contexts, challenging the uniform implementation approaches common in practice.

For practitioners navigating this emerging technological landscape, our framework provides actionable guidance for strategic positioning, implementation approach, and consumer experience design – enabling more effective value creation and appropriation from voice technology investments. As voice interfaces increasingly mediate the customer–brand relationship, understanding these nuanced dynamics becomes essential for sustainable competitive advantage in the AI era.

2. Literature review

Artificial Intelligence refers primarily to a system’s “ability to behave in a way that could be considered intelligent if it were a human being” (McCarthy, 2007). AI-enabled voice agents/assistants (VAI) are digital applications employing machine learning and artificial intelligence to mimic human-like conversations. Artificial intelligence in voice can improve task performance (Lopatovska *et al.*, 2019), fostering positive attitudes toward VAIs (Moriuchi, 2019). As Voice AI becomes a tool for improving business reach and efficiency, organizations explore integration methods to trigger desired customer outcomes.

2.1 Adoption and use of voice AI

Most studies identify PEOU (perceived ease of use) and PU (perceived usefulness) as significant drivers of intention to adopt (Venkatesh and Davis, 2000; Venkatesh *et al.*, 2003; Moriuchi, 2019) and use intelligent voice assistants among consumers (Yang and Lee, 2019; Terzis *et al.*, 2012), driving consumer engagement and loyalty (Pal *et al.*, 2020). However, Kowalczyk (2018) found that ‘perceived ease of use’ lacks a significant impact on behavioral intention or ‘perceived usefulness’. The nonvisual nature of voice agents and difficulty comparing products leads consumers to defer purchases (Mueller *et al.*, 2017).

Beyond functional attributes triggering VAI adoption, the literature examines variables ensuring continued use and acceptance, with particular attention to drivers of consumer trust and attitudes toward VAI interactions. Trust positively moderates the relationship between expectation confirmation and customer satisfaction (Brill *et al.*, 2022). Innovation Resistance Theory (Ram and Sheth, 1989) discusses consumers’ resistance stemming from potential changes to the satisfactory status quo or conflicts with belief structures. While Voice AI increases work efficiency, uncertainties surrounding its use act as adoption barriers.

Voice technology helps build emotional connections with brands, allowing marketers to adopt unique brand voices creating personas that better connect with consumers. These emotional connections influence information processing and product evaluation (Jones, 2018; Singh, 2022; Prentice *et al.*, 2023). Task nature also determines Voice AI use. Utilitarian benefits motivate VAI use as it requires lower trust and assurance levels (Fernandes and Oliveira, 2021; Pitardi and Marriott, 2021). McLean and Osei-Frimpong (2019) found that social, utilitarian and symbolic benefits motivate VA usage, while hedonic benefits do not. Privacy concerns demotivate users. Bawack *et al.* (2021) proposed that in

voice shopping contexts, privacy concerns affect the intention to use voice shopping services but not the customer's overall experience. While these adoption factors establish the foundation for VAI implementation, understanding the variables that drive ongoing engagement is crucial for sustaining value creation beyond initial acceptance.

2.2 Variables impacting engagement with voice AI

Consumer engagement proves crucial for creating loyalty toward voice assistants for both transactional and nontransactional VAI use. Localization and language customization were important for nontransactional relationships (Moriuchi, 2019). Poushneh (2021) demonstrated that VAI personality traits drive voice interaction flow experience, resulting in satisfactory experiences. Consumers feel controlled when interacting with VAI, displaying functional intelligence, sincerity and creativity.

Beyond trust and engagement (Moriuchi, 2021), anthropomorphism emerges as an important theme, with attention to its role across consumer interaction stages. VAI mimic human-like attributes that elicit social presence and infer social cognition, nurturing trustworthy relationships (Choi and Drumwright, 2021). Rhee and Choi (2020) posited that personalized content and the social role of a friend lead to more positive attitudes toward products with voice assistants. Anthropomorphic cues may have negative consequences like impairing self-disclosure and evoking socially desirable responses (Schuetzler *et al.*, 2019; Pickard and Roster, 2020).

Having established the factors influencing consumer engagement with VAI, we now examine how these interactions translate into tangible value creation for firms.

2.3 Value creation using voice AI

Bahmani *et al.* (2022) found that some firms experience valuation increases after launching voice assistants, while others show no change or decreases. The market response is influenced by voice assistant features and firms' business nature. Researchers measured different VAI capability values and the impact of voice assistant feature announcements on firm value. Mari *et al.* (2020) examined managers' perceptions regarding shopping-related VAI evolution and potential effects on marketing practice. Companies need to prioritize strategic investments involving voice ecosystem building, tactical investments in voice search, and people training. As consumers show increased dependency and preference for voice assistants, brands must redesign their value chain and value proposition. Table 1 summarizes the academic literature.

While academic literature has begun to explore VAI's value creation potential, examining industry perspectives provides complementary insights into real-world implementation challenges and opportunities.

2.4 The trade literature on voice AI

A recurring theme across firm-side literature has been voice AI utilization to enhance customer service experiences, and engagement. Gallemard (2020) emphasized voice AI's potential to reduce customer service costs while improving interaction quality. Managerial literature also addresses return on investment (ROI) from VAI, with the PWC and IMAI (2022) report highlighting higher returns and speedy cost recovery.

As voice assistants become more context-driven and proactive rather than merely reactive, firms must test how voice technology fits their business while addressing privacy and security concerns (Metrock, 2019). The industry emphasizes strategic Voice AI adoption (Dawar and Bendle, 2018), advising firms to develop strategies to manage consumer expectations effectively. Financial Express (2019) and PWC (2018) report emphasize that

Table 1. Academic literature – summary

S. No.	Author	Key insights	Consumer focus	Firm focus	Monetization focus
1	Aw <i>et al.</i> (2022), Malodia <i>et al.</i> (2022), Malodia <i>et al.</i> (2023), Fernandes and Oliveira (2021), Kowalczuk (2018), Choi and Drumwright (2021)	Consumer adoption of digital VA, factors impacting the acceptance/avoidance of use and continuance of use. Study the effect of constructs like perceived usefulness (PU), perceived ease of use (PEU), social norms and utilitarian/hedonic motivations	Yes		
2	Guha <i>et al.</i> (2023)	Lower levels of perceived voice assistant (VA) artificiality and higher levels of perceived VA intelligence increase consumers' VA evaluations, moderated by users' perceived sacrifice, length of VA ownership and age	Yes		
3	Moriuchi (2019)	Consumer engagement via the technology serve as a partial mediator for the relationship between perceived use and loyalty for both transactional and nontransactional activity	Yes		
4	Hernandez-Ortega and Ferreira (2021)	Frequent interactions and positive emotions turn users-VA exchanges into affective relationships	Yes		
5	Singh (2022)	Voice Assistant (VA) anthropomorphism and trust in VA predicts consumer–VA emotional attachment and establishes the moderating role of consumer self-disclosure	Yes		
6	Pitardi and Marriott (2021)	VA mimic human-like attributes through voice communication that elicit a sense of social presence and inferences of social cognition in the mind of the user that, in turn, nurture trustworthy relationships	Yes		
7	Pagani <i>et al.</i> (2019)	Impact of privacy concern on the relationship between personal engagement and trust depends on the nature of the platform interaction	Yes	Partially	

(continued)

Table 1. Continued

S. No.	Author	Key insights	Consumer focus	Firm focus	Monetization focus
8	Jain et al. (2022)	Brand credibility moderates the relationship between voice assistant features and the overall perceived value. Utility and hedonic features contribute to the overall perceived value of voice assistants	Yes		
9	Vernuccio et al. (2023)	The design of human-like brand voice characteristics like dialogue features (i.e. intuitiveness, responsiveness and continuous adaptation) aims to strengthen perception of a brand personality and the development of a stronger relationship should be geared towards increasing brand loyalty		Yes	Partially
10	Grewal et al. (2022) , Klaus and Zaichkowsky (2022)	Modifies traditional consumer decision-making process to integrate technology and artificial intelligence. VA evaluations include perceptions of humanness and artificiality in customer journey	Yes		
11	Bawack et al. (2021)	Privacy concerns had no direct effect on customer experience performance but affect intention to use and actual use of voice shopping services. Trust mediates the relationship between privacy concerns and customer experience performance. Prior experience positively affects consumer trust. Personality differences affect perceptions of customer experience	Yes		
12	Rabassa et al. (2022)	Conversational commerce produces dark side effects that limit its appropriation, mismatching expectations and preferences. Customers feel constrained in final choice making	Yes		Partially
(continued)					

Table 1. Continued

S. No.	Author	Key insights	Consumer focus	Firm focus	Monetization focus
13	Hu et al. (2023)	Incongruity between the speaking and listening attributes undermine consumers' trust in voice assistants, leading to reduced willingness to accept product recommendations	Yes		
14	Hu et al. (2022) , Rhee and Choi, (2020)	When users perceive power over AI assistants, they feel AI assistants' behaviors are under their control. Power experiences attenuate users' risk perceptions regarding consumption. Personalization and the social role play help build attitudes toward the product	Yes		
15	Bahmani et al. (2022)	Measured the value of different types of capabilities of VAI, and the impact of announcements of VAI features on firm value		Yes	Yes
16	Halbauer and Klarmann (2022)	Customer's good mood is associated with purchases of higher-priced premium brands. Retailers can use this to adapt the presentation of product, to help customers optimize their decisions	Yes	Partially	
17	Mourey et al. (2017)	Interaction with anthropomorphic consumer products (through design, interaction, intelligence, responsiveness and/or personality) can satisfy social needs, and mitigate consumer social exclusion	Yes		
18	Melzner et al. (2023)	Introduces mechanisms in oral versus visual consumer expression and integrates them into a comprehensive decision-making framework to illustrate how they can influence verbal disclosure	Yes	Partially	

(continued)

Table 1. Continued

S. No.	Author	Key insights	Consumer focus	Firm focus	Monetization focus
19	Crolic et al. (2022)	In chatbot-led service interaction with customer in an angry emotional state, chatbot anthropomorphism has a negative effect on customer satisfaction, firm evaluation, and purchase intentions	Yes		
20	Luo et al. (2019)	Disclosure of chatbot identity before the machine–customer conversation reduces purchase rates by more than 79.7%. When customers know the conversational partner is not a human, they are curt and purchase less	Yes		
21	Mari et al. (2020)	Manager’s usage of voice commerce is correlated with their trust towards VAI, perception of personalization and consumer’s intention to delegate the buying process to the VAI. They view of the importance of voice commerce for their industry and the opportunity for their brand		Yes	Partially
22	Wamba-Taguimdje et al. (2020)	Analyze influence of AI on firm performance, and build on the business value of AI-based transformation projects. AI expresses its potential through its ability to optimize existing processes and improve automation, information and transformation effects		Yes	Partially
23	Dellaert et al. (2020)	Conceptual work on consumer behavior with use of VAI	Yes	Partially	
24	Kim and Lee (2016)	User experience positively moderates the relationship between uncertainty costs and user resistance. It negatively moderates the association between perceived value and user resistance	Yes		

(continued)

Table 1. Continued

S. No.	Author	Key insights	Consumer focus	Firm focus	Monetization focus
25	Mishra <i>et al.</i> (2022)	Playfulness and escapism positively influence hedonic attitude. Anthropomorphism and social presence determine utilitarian attitude. Utilitarian attitude has a stronger impact on usage and WOM recommendations	Yes		
26	Breidbach and Maglio (2016)	Theoretical contribution includes five propositions including actors, resources and practices underlying technology-enabled value co-creation in B2B service systems	Yes	Yes	
27	Kohtamäki and Rajala (2016)	Value co-creation and the coproduction of value proposition covers a myriad of viewpoints to the economic and social exchange among actors in multi-actor B2B service ecosystems		Yes	
28	Moradi and Dass (2022)	How B2B marketers perceive the benefits of AI adoption. Discusses the perceived barriers to AI adoption, data privacy challenges and the replacement of human workforces	Partially	Yes	
29	Our paper	How VA firms can use B2B and B2C markets to monetize and create value for business	Partially	Yes	Yes

Source(s): Authors' own work

brands must adopt voice AI to maintain competitiveness, especially as it becomes integral to the customer journey, influencing acquisition, satisfaction and retention. Future predictions highlight the importance of seamless information flow and competitive advantages for early adopters (Marr, 2022).

This managerial perspective highlights the strategic importance of VAI adoption, yet it must be considered within the broader context of integrated B2B and B2C marketplaces.

Academic literature on voice assistants has concentrated on understanding customer adoption and use factors, with dominant focus on perceived benefits and ease of use (Venkatesh and Davis, 2000). Managerial literature maintains a business perspective,

recognizing the importance of integrating voice technology in business functions. While existing theoretical frameworks provide valuable foundations for understanding technology adoption and value creation, they prove fundamentally inadequate for explaining the complex dynamics we observe in VAI implementation. This inadequacy stems not from minor theoretical gaps that could be addressed through framework extension, but from core assumptions that are contradicted by emerging VAI implementation patterns.

The counterintuitive relationships, bidirectional marketplace effects, context-dependent technology characteristics and regulatory override mechanisms in VAI represent fundamental departures from established theoretical assumptions rather than incremental variations that could be accommodated through framework modifications.

2.5 Integrating B2B and B2C marketplace

The Technology Acceptance Model (Davis, 1989) and its extensions, including UTAUT (Venkatesh *et al.*, 2003), conceptualize technology adoption as primarily user-driven processes where perceived ease of use and perceived usefulness directly predict adoption intentions. However, our preliminary investigations reveal that VAI exhibits counterintuitive adoption patterns that fundamentally challenge these assumptions.

VAI implementation requires simultaneous consideration of B2B and B2C adoption dynamics, where end-user comfort creates “pull” effects that influence organizational investment decisions. Traditional acceptance models cannot accommodate this bidirectional marketplace interdependence, as they focus on individual-level adoption rather than the complex ecosystem effects, characteristic of VAI deployment. Ramani and Kumar’s (2008) Interaction Orientation Framework suggests that technological advances increase interaction opportunities between firms and customers, between customers, and between firms. Interaction orientation reflects a firm’s ability to interact with customers and use information to achieve profitable relationships. Three actors create this value: the technology provider, the deploying firm and the consumer (Kot and Leszczyński, 2022). In VAI applications, value creation and appropriation depend on end-user (B2C) adoption, giving B2B customers the confidence to invest in VAI applications (Figure 1) to create better customer experience and greater business efficiency.

As evident in Tables 1 and 2, clarity exists on how firms can use VAI, but a gap remains in addressing factors influencing value creation and monetization mechanisms. This gap

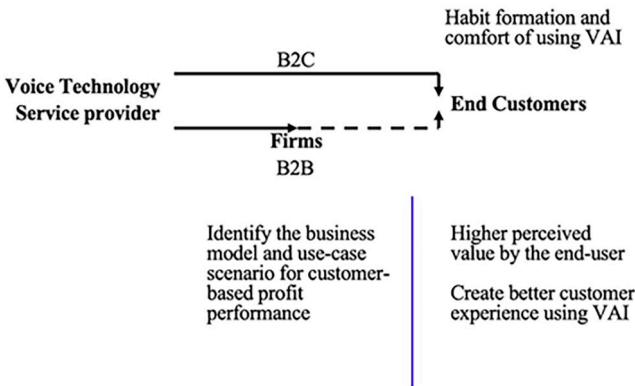


Figure 1. Integration of B2B and B2C marketplaces for VAI value creation process

Source: Created by authors

Table 2. Marketplace evidence – summary

S. No.	Author	Key insights	Consumer focus	Firm focus	Monetization focus
1	Gallemand (2020)	Use of voice chatbots to enhance customer service experience and decreases cost of customer contact	–	Yes	Yes
2	Business Insider (2018)	Use of VAI by retailers to achieve sales gains of 6–10%	–	Yes	Yes
3	Metrock (2019)	Need for strategy to deploy voice AI and manage its capabilities	Partially	Yes	Partially
4	Think with Google (2020)	Use case of voice by consumers and importance of vernacular capabilities	Yes	Partially	–
5	Dawar and Bendle (2018)	Voice tech will impact firms at three critical levels: customer acquisition, satisfaction and retention. Consumer loyalty will be towards AI assistant and not brands	Yes	Yes	Yes
6	PWC (2018). Report	Search, advertising, content and commerce will impact consumers transform the way they interact with brands	Yes	–	–
7	PWC and IAMAI (2022)	The return on investment for voice AI is very high and the initial cost is typically recovered within the first few months of implementation	–	Yes	Yes
8	Capgemini (2019) Report	Convenience and ability to do things ‘hands free’ by using VAI. Firms see 25 points NPS ® advantage among voice assistant users for a brand	Yes	Partially	Partially
9	Vixen Labs (2022)	Voice AI has clear alignment with the marketing funnel from awareness through purchase and retention	Yes	–	–
10	Mayer and Harrision (2019)	Significant share of commerce will be carried out by a generation used to shopping by voice. The adoption curve will be much steeper	Yes	Partially	–
11	Simms (2019)	Voice can help in decision making and almost becomes an influencer in process	Yes	Partially	–
12	Michelman (2017)	How brand communications and marketing strategies should evolve in the age of voice	Yes	Partially	–

(continued)

Table 2. Continued

S. No.	Author	Key insights	Consumer focus	Firm focus	Monetization focus
13	Mee and Ozturk (2020)	Cyberattacks on voice-recognition systems with greater use of VAI	Yes	–	–
14	Kumar <i>et al.</i> (2019)	VAI can provide predictions regarding the AI-driven environment on branding and customer management practices	Yes	Yes	–
15	Gupta (2022)	Voice AI can create an information bridge where conversations flow seamlessly. Better CX, lower cost and scalable reach to develop a competitive edge	Partially	Yes	Partially
16	Goorha and Iyengar (2022)	Advanced analytics on data generated from customer to reduce average handle time by 40%, increase self-service containment rates by 5–20%, cut employee costs by up to \$5 mn, and boost the conversion rate on service-to-sales calls by 50%	–	Yes	Yes
17	Hyken (2017)	Personalized experience through voice enhances loyalty and revenue	Partially	Yes	Yes
18	Our paper	How VA firms can use B2B and B2C markets to monetize and create value for business	Partially	Yes	Yes

Source(s): Authors' own work

informs our focus in the remainder of this research, where we attempt to answer how firms integrate VAI across B2B and B2C marketplaces to create and appropriate value with VAI.

3. Method

Grounded theory methodology provides the appropriate foundation for developing new theoretical understanding of these complex, emerging phenomena. By systematically examining VAI implementation practices across multiple organizational contexts and stakeholder perspectives, we can develop middle-range theory that accurately represents the unique dynamics of voice technology value creation and appropriation while contributing to broader understanding of AI-enabled technology implementation in interconnected marketplace environments.

Our research employs a rigorous grounded theory approach with methodological triangulation to develop a robust theoretical framework. Beginning with a comprehensive gap analysis of academic and marketplace literature, we systematically conducted three interconnected studies to generate and validate theory around VAI implementation and value

creation. This methodological approach enables us to move beyond descriptive accounts of VAI adoption to develop robust explanations of value creation and appropriation mechanisms that address the critical theoretical gaps while providing actionable guidance for organizations navigating this emerging technological landscape. The iterative nature of grounded theory – with its constant comparison between emerging concepts and new data – allowed us to identify counterintuitive relationships and contextual factors that might be overlooked in hypothesis-testing approaches.

The grounded theory methodology (Glaser and Strauss, 1967) provides an ideal foundation for this research, as it excels in developing theoretical frameworks for complex, emerging phenomena like VAI where existing theory is insufficient. Following established grounded theory protocols, we collected rich qualitative data through multiple complementary studies, systematically analyzed this data through iterative coding procedures and developed theoretical constructs that accurately represent real-world practices.

Study 1 captured essential consumer perspectives through a semistructured online survey with 197 participants aged 25–45 years recruited via Qualtrics. This age demographic represents the primary VAI adoption segment based on industry adoption data (Vixen Labs, 2022), ensuring our foundational understanding captured perspectives from users most likely to drive B2C marketplace dynamics that in turn influence B2B adoption decisions.

The sample size follows established guidelines for exploratory research in emerging technology contexts, where the goal is pattern identification rather than population parameter estimation (Hair *et al.*, 2019). Respondent profiles (detailed in Supplementary Table 1) ensured appropriate representation across demographic variables including income, education, technology usage patterns and geographic distribution within our target demographics. This survey provided crucial insights into voice assistant usage, adoption barriers and engagement triggers – establishing the end-user foundation necessary for understanding value creation potential.

Study 2 expanded our investigation to the organizational level through 18 structured focus groups with business professionals. Each group comprised seven to eight executive education participants from diverse sectors, representing various organizational roles and implementation experiences. These discussions, conducted at a neutral location using consistent semistructured protocols, revealed organizational perspectives on VAI implementation challenges and opportunities. We systematically analyzed transcripts to identify emerging themes, validating them through subsequent focus groups – a key procedural element of rigorous grounded theory development. Our sampling strategy evolved through three distinct phases aligned with grounded theory's theoretical sampling principles. Study 1 employed purposive sampling targeting primary VAI users to establish foundational understanding. Study 2 used convenience sampling from executive education participants, providing diverse organizational perspectives while maintaining accessibility for structured discussions. Study 1 and Study 2 aided triangulation approach and helps to proceed to Study 3, which is based on principles of grounded theory.

Study 3 represented the culmination of our theory-building process through in-depth expert interviews. It employed theoretical sampling, where participant selection was guided by emerging theoretical categories from previous studies. We identified 23 experts across sectors, with seventeen ultimately participating in 90 to 120 minute interviews. Participants held direct decision-making authority for VAI implementation, with experience ranging from seven to thirty-one years (profiles in Supplementary Table 2). Our participant selection criteria ensured representation across:

- diverse organizational contexts (size, sector, industry position);
- both B2B and B2C market orientations; and

- various stages of VAI exploration, implementation and integration.

This theoretical sampling continued until reaching saturation, where additional interviews yielded no new insights – a standard criterion in grounded theory research.

The extensive interviews followed established grounded theory protocols with two distinct phases: establishing organizational context, then exploring specific VAI applications, implementation processes and value creation mechanisms. All interviews underwent recording, *verbatim* transcription, and systematic coding. We conducted interviews in small batches, analyzing each set before proceeding to the next, until reaching theoretical saturation where no new constructs or relationships emerged, confirming comprehensive theory development.

Data analysis followed [Corbin and Strauss's \(1990\)](#) systematic three-stage coding methodology, using NVivo software to ensure analytical rigor. Two researchers independently conducted parallel coding processes to enhance reliability. In open coding, we identified 56 distinct conceptual categories from the raw data. During axial coding, we established relationships between these categories, consolidating them into 11 first-order constructs. Through selective coding, we further integrated these into six core theoretical dimensions that form the foundation of our framework. Supplementary Table 3 provides the complete coding structure, illustrating clear connections between raw data and theoretical constructs – a hallmark of methodologically sound grounded theory. [Figure 2](#) gives an overview of the method.

3.1 Reliability and validity assessment

We implemented multiple methodological safeguards to ensure the reliability and validity of our theoretical framework. First, our multi-method design provided inherent triangulation,

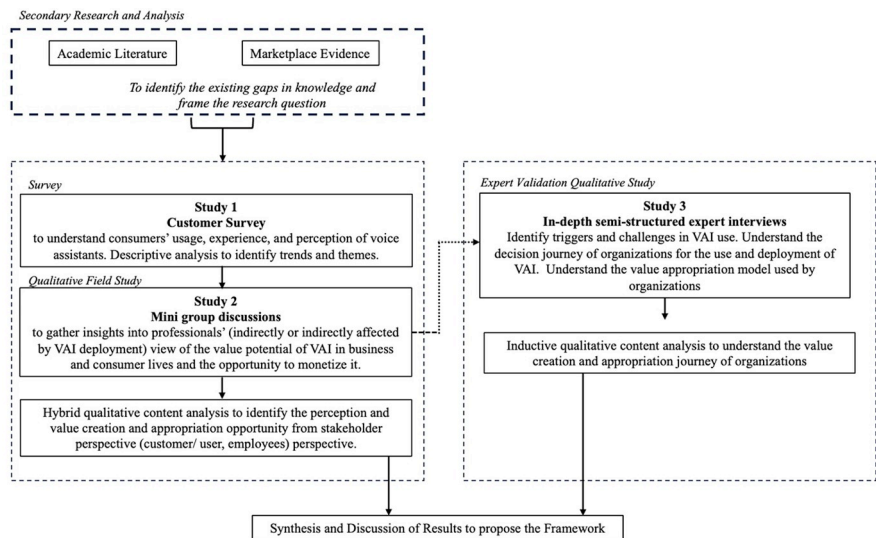


Figure 2. Illustration of research method
Source: Created by authors

with each study validating and extending insights from previous phases. The progression from consumer survey to professional focus groups to expert interviews created a comprehensive, multi-perspective foundation for theory development.

To ensure coding reliability, two researchers independently analyzed the data using identical protocols. Their parallel coding structures underwent comparative analysis to establish consensus on emerging theoretical constructs. Additionally, two independent scholars with expertise in qualitative methods but no prior involvement in the research reviewed our coding methodology, analytical process and theoretical categorizations – confirming procedural rigor through investigator triangulation (Bell and Bryman, 2007).

We established validity through member checking, sharing framework summaries with eight study participants who confirmed the accuracy of our conceptualization with minor terminological refinements (Guba and Lincoln, 1994), and that our theoretical constructs authentically represented participants' experiences and industry practices.

Finally, we subjected our framework to critical examination through presentations to both academic and practitioner audiences, incorporating their feedback to refine theoretical constructs and relationships. Through this methodically rigorous approach to grounded theory development, we established a theoretically sound and empirically validated framework for understanding VAI value creation and appropriation mechanisms – addressing critical gaps in existing literature while providing actionable insights for practitioners navigating this emerging technological landscape.

4. Results

4.1 Study 1: User survey

With the significant growth in voice technologies usage and younger populations in emerging markets, understanding such markets is a managerial priority. Our user survey revealed four key dimensions of VAI usage patterns and perceptions:

4.1.1 Primary use cases. The most common tasks for using voice assistants are playing music, setting reminders, placing calls, controlling smart home devices, getting weather updates and playing games. Beyond these basic functions, VAI was also used to search for information, compare products, access specialized content and place orders.

4.1.2 E-Commerce integration. Opinions on voice-based e-commerce were divided. Some participants found it convenient for routine purchases, while others expressed strong preferences for more control, visual comparison capabilities and access to reviews before making purchase decisions.

4.1.3 Adoption advantages. Respondents highlighted several distinct advantages of voice assistants. They reported that VAs could be used by people of all ages, literate or not, with minimal effort. Unlike other technologies, VAI demonstrates a remarkably flat learning curve as voice encompasses natural prompts that deliver comprehensible responses to most users. Key positive aspects included ease of use, hands-free functionality and time-saving capabilities.

4.1.4 Adoption barriers. Despite the advantages, participants identified several concerns that limit broader adoption. Speed and accuracy issues, privacy concerns and inconsistent device integration emerged as primary barriers to more extensive VAI usage.

This systematic assessment of consumer perspectives helped establish the foundation for understanding VAI's perceived value and potential use cases, which informed our subsequent investigations with business stakeholders. Supplementary Table 4 provides a comprehensive summary of customers' negative and positive sentiments regarding VAI.

4.2 Study 2: Managerial mini-group discussions

Building on our understanding of user sentiment, we mapped the VAI landscape from a managerial perspective through structured mini-focus group discussions with professionals in an executive program at a leading business school in India. These discussions focused on two core objectives:

- (1) identifying potential VAI use cases from the perspective of represented businesses; and
- (2) characterizing the essential attributes of each use case.

Using inductive coding procedures (Kaplan and Maxwell, 2005), we systematically analyzed the resulting data to identify emergent themes.

4.2.1 Strategic value creation. The first major theme that emerged centered on strategic applications – ways firms could leverage VAI to gain sustainable competitive advantages. Participants identified two primary strategic pathways:

- (1) *Customer empowerment:* Applications focused on enhancing customer capabilities, reducing friction points and delivering superior experiences through voice-enabled interfaces.
- (2) *Business process enhancement:* Applications targeted at improving internal efficiencies, reducing operational costs and enabling more effective decision-making processes.

4.2.2 Monetization approaches. The second major theme focused on subscription and monetization mechanisms – specific ways firms could capture economic value from VAI implementation. Participants distinguished between:

- *B2C monetization models:* Direct revenue generation through consumer-facing applications, premium features and enhanced services.
- *B2B monetization models:* Enterprise value creation through efficiency gains, licensing opportunities and integration with existing business solutions.

Supplementary Table 5 provides a detailed mapping of the various strategies and applications associated with these themes. This managerial perspective established that firms envision benefiting from VAI through both value creation and direct monetization, but raised two critical questions requiring further investigation: (a) which specific pathways offer the greatest potential for value creation and appropriation; and (b) which subscription models will prove most effective in various implementation contexts. These questions guided our design of Study 3.

4.3 Study 3: Expert interviews

In the expert interviews (Von Soest, 2023), it became evident that apart from PU and PEOU, many other variables play a role in firms' decisions regarding investment in VAI. These include scaling up variables, which help organizations create value for customers and enable them to monetize. The “decision to invest” is the central theme of our framework and measures the level of investment the firm is willing to make, which further determines the business model it chooses. Finally, the business value created, includes the direct and indirect method of monetizing the value created for customers using VAI. Table 3 describes the results of expert interviews. Based on our three studies, we propose a framework to help companies make decisions regarding value creation and monetization with VAI investments.

Table 3. Summary of results: expert interviews (Study 3)

Themes	Variables	Exemplar from Interviews (Study 3)	Variables from theory explaining the variables	Definitions
Bottlenecks	Challenges of VAI adoption	E-commerce is a common use case in VAI, and people are aware of it. But they never wanted to go with them because they are scared about their credit card. Details or reluctant to order without seeing-it. (ID16)	Usage and value barriers from innovation resistance theory	The challenges in the adoption of the B2B and B2C marketplace, which forms a base for value creation using VAI
<i>Main effects</i> Business value created	Value creation with VAI	AI enabled voice technology can help us in invisible operations or human free operations. Our goal is to replace our call centers largely with AI voice bots and this is having a bottom-line impact on call center operations. ID10 Content building using AI-enabled voice technology will require regional language adoption and can help in new customer acquisition. Regional brands will benefit a lot in Tier 3 towns as that is their core market. Engagement with voice assistants will lead to new trials. Retention will depend on cart conversions. ID1	Customer-based relational performance and profit performance (interaction orientation theory)	Value appropriation through increased profits/ decreased costs/ referrals
Customer-related factors	B2C market use cases	My son asks Alexa to recite him a bedtime story every day! The young generation is more comfortable with the use of VAI. They make purchases, pay bill, and can trust their VA to manage life. For them, it will be easy to interact with a voice	Customer empowerment (interaction orientation theory)	Developing an ecosystem for design and delivery of the best VAI solution

(continued)

Table 3. Continued

Themes	Variables	Exemplar from Interviews (Study 3)	Variables from theory explaining the variables	Definitions
Decision to invest	Customer pull	bot while booking a hotel or buying insurance. They have grown with it. ID4 If I use voice technology as a personal assistant, it has access to my travel plans and my calendar, and what if someone hacks all my information? ID9 Pull power of VAI by end users is what firms can bank on. If the customers of businesses are ready to try out this new technology, then only it is going to be a success. If a greater number of people use these devices, a lot more adoption will happen. They will then seek smarter, intelligent devices at home and switch to voice commands for many applications. ID1	Customer concept (interaction orientation theory)	Customer being the center and if they accept the VAI, it gives confidence to business to use it
	Investment needed	When I buy or license, the third party will keep monitoring it and working on upgrades needed. I am hence sorted with collaborating with them for application. ID14 We must look at a scale because if the scale is very large, then their cost will be lesser. If the scale is lesser, then the cost will be higher. So, it will depend on the	Business value management, customer empowerment (interaction orientation theory)	Choice of business models based on willingness to invest (based on antecedents and moderators)

(continued)

Table 3. Continued

Themes	Variables	Exemplar from Interviews (Study 3)	Variables from theory explaining the variables	Definitions
	Evolution of technology	roadmap with respect to the scale which they have and the timelines they have set. ID14 For greater efficiency, one will have to invest in somebody who is improving VAI over time. So, partnership arrangement where someone can maintain the platform for you, but you own the data is going to be critical. ID7		Choice of business models requires collaboration and continuous feedback mechanism to deploy, deliver and improve over time
VAI Deployment – Triggers	Triggers to deploying VAI	Adoption of VAI from a customer’s perspective, either from a text to voice vs. human beings, depends on the importance of the problem they are trying to solve. ID5 Not everyone can do this! Voice technology can give a differentiated customer experience. This differentiation is needed for profitability. But it involves extensive capital investment as training for people and machines is needed. ID10 Muscle memory will also make a lot of difference. Your experiences become your knowledge over time. Now, prior success need not be only on VAI, but any AI technology affects our investment. ID6	Usage barrier (innovation resistance theory) Value barrier resolution (innovation resistance theory) Usage barrier resolution (innovation resistance theory)	

(continued)

Table 3. Continued

Themes	Variables	Exemplar from Interviews (Study 3)	Variables from theory explaining the variables	Definitions
Adoption and scale variables	Past success with technology	Quick wins help. So, one thing certainly is that if you have something that's worked for so long and it is kind of mature and stable, then you don't want to experiment with new stuff to build a competitive advantage. Inertia always comes into play with a lack of experience in AI-enabled technology. ID13	Usage barrier resolution (innovation resistance theory)	
	Customization	I can use Alexa skills to offer personalization for my guests. They can track their history, and the device will know what coffee my guest wants or if they have special needs. My job is done! I have Alexa skills readily available at less cost. ID9	Interaction response (interaction orientation theory)	Ability to service heterogenous customers and fulfil their customized needs
	Application specificity	In case of Fintech, where the product is extremely difficult for the consumer to understand, you can use voice technology to sell the product. Then ROI is measurable. There are standard application that do this job. In the chocolate market, for the next couple of years, we may not have anything that requires VAI, we wouldn't want to invest heavily in building specific application. ID12		

(continued)

Table 3. Continued

Themes	Variables	Exemplar from Interviews (Study 3)	Variables from theory explaining the variables	Definitions
	Regulatory environment	The banking industry is regulated and ring-fenced, so firms cannot cross it by use of technology. Otherwise, they will get hurt. We ask for safer and established applications from service providers. ID6	Risk barrier resolution (innovation resistance theory)	Firms operating under regulatory restrictions have more barriers to adoption of new technology

Source(s): Authors' own work

5. Discussion

5.1 The framework

Our theoretical framework represents a significant advancement beyond existing models by challenging three prevailing assumptions in the literature. First, traditional technology acceptance models (Davis, 1989; Venkatesh et al., 2003) conceptualize adoption as primarily user-driven. Our framework reveals the complex bidirectional influences between B2B and B2C marketplaces in VAI implementation, where B2C creates a pull for B2B marketplace to adopt VAI. This pull is critical for monetization success of VAI. Second, whereas existing value creation literature often presumes linear relationships between investment intensity and value appropriation (Amit and Zott, 2001), our research uncovers counterintuitive nonlinear dynamics, where higher integration potential leads to lower investment intensity. Third, our framework extends beyond the technology–organization–environment (TOE) perspective (Tomatzky and Fleischer, 1990) by demonstrating how regulatory constraints can fundamentally alter organizational responses to otherwise compelling business opportunities. These limitations in established frameworks necessitated our grounded theory approach, as modification of existing models could not accommodate the complex, counterintuitive relationships .

Based on our study, we propose a framework for “Value Creation and Appropriation using Intelligent Voice Technology” (Figure 3). While existing approaches often treat VAI adoption as a uniform process, our research reveals a more nuanced reality governed by counterintuitive relationships and context-dependent variables. The literature establishes that firm-customer relationships yield positive implications for brand and customer experience (Aw et al., 2022; Hernandez-Ortega and Ferreira, 2021). Opportunities for firms to create customer value extend marketing practices into customers’ spheres through firm-customer interactions (Grönroos, 2011).

Using propositions from the Interaction Orientation Approach (Ramani and Kumar, 2008), firms can model customer–firm relationships around:

- adopting VAI into use-cases that create value for firms and consumers;
- choosing a business model suited to organizational context to enhance value created for both B2B and B2C marketplaces; and
- leveraging end-customer comfort with VAI to create “pull” power that motivates firms to adopt VAI across multiple applications.

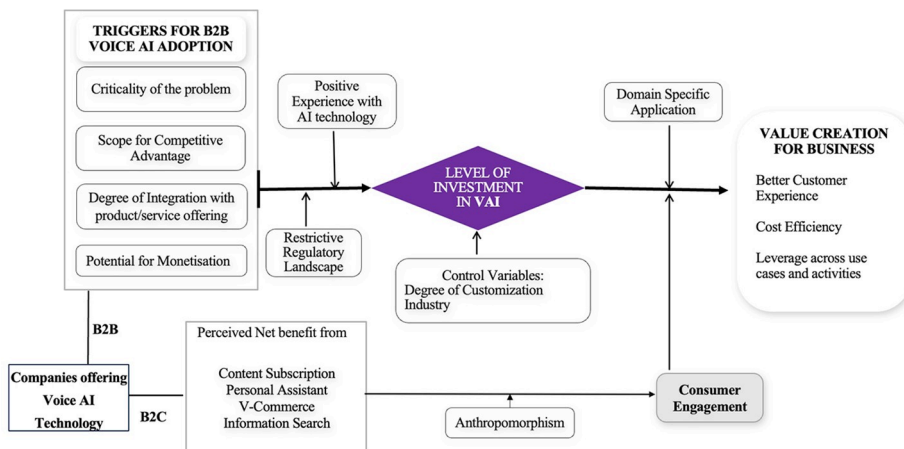


Figure 3. Value creation and appropriation using voice AI technology (VAI): integrating B2B and B2C marketplaces

Source: Created by authors

Our framework suggests two paths for value creation using VAI – B2B and B2C marketplace approaches. The B2B path variables divide into three themes:

- (1) triggers of VAI adoption;
- (2) business model based on VAI investment level; and
- (3) value created for business.

The B2C marketplace themes include the following:

- perceived net benefit in possible use cases; and
- customer engagement. Table 4 presents the variables with their conceptual and operational definitions.

The firm's investment level determines whether it builds its own VAI platform or buys services from providers (Figure 4). Our results and discussions lead us to propose testable propositions.

5.2 Propositions

5.2.1 Degree of product integration and its effect on the level of voice AI investments.

Firms adopt new technology through user pull or competitive push. We discovered a striking paradox in adoption motivations: they place asymmetric demands on investments in ways that contradict typical technology implementation patterns suggested in Technology Acceptance Model and its iterations (Davis, 1989, Venkatesh *et al.*, 2003). These models provide no mechanism for understanding how competitive standardization and outsourcing might reverse this relationship. When users demand voice-based interactions, they expect a flat learning curve. As firm offerings integrate seamlessly with VAI, competitors also find incorporating VAI solutions easy. This enforces replication or standardization, depresses innovation or distinction, and drives firms to seek subscriptions, white-label solutions, or

Table 4. Variables in the framework and their definitions and operationalization (Study 3)

Variables	Definitions	Operationalization
<i>B2B triggers for VAI adoption</i>		
Criticality of the problem	Criticality is measured with respect to its importance in delivering the firms' offerings and create desired value for customers and business	Measure of 'how critical is the problem'. The scale of 1–3 measures the criticality with 1 – not so critical, 2 – moderately critical, 3 – highly critical
Scope for competitive advantage	Measure of how VAI can help the firm to differentiate itself from competition	The three-point scale would measure low potential to differentiate – to – high potential to differentiate
Degree of integration with product/service offerings	Measure of ease with which VAI can be integrated with organizations' product/service offerings	The three-point scale would measure degree of integration with VAI
Potential for monetization	Measure of the price premium that customers will tolerate when VAI-based product/solution addresses a pain-point or a gap	The three-point scale would measure potential for monetization. (low, medium, high)
<i>Moderating variables</i>		
Regulatory landscape	Measure of how stringent or relaxed is the regulatory environment governing the sector, which will impact the extent to which firms can use VAI	Measure using 0–1 scale; with 1 being a restrictive regulatory landscape and 0 being relaxed regulations
Positive experience with AI technology	Orientation of the firm to deploy new technology and absorb risks of adoption. Firm's tech disposition could be impacted by its past success with AI technology, risk appetite for new technology, and willingness to be "ahead of the curve"	1–3 Scale measures the past experience with AI technology. 1-negative experience, 2-neutral and 3-positive experience
Domain-specific application	Measure of specialized nature of the VAI application and its ability to carry out sector/ use specific tasks with higher level of efficiency	Measured on a three-point scale. 1 – generic application 2 – moderately specialized. 3 – highly specialized
Customer engagement	Holistic measure of customer's cognitive, affective, emotional, social and physical responses to the entity, product and service (Verhoef <i>et al.</i> , 2009). Customer contribution or engagement is reflected in customer purchases, customer referrals, customer influence and customer knowledge (Kumar <i>et al.</i> , 2010)	Kumar and Pansari (2016) use a 5-point Likert scale with 16 items to measure customer engagement. The reliability of the scale in their study exceeds 0.8
<i>Central theme – main effects</i>		
Level of investment in VAI	The measure of level of investment the firm is ready to make in VAI, which determines the business model chosen	1–3 scale measuring low to high levels of investment
<i>Outcome variable: value creation for business</i>		
Cost efficiency	Extend to which the cost (firms' offering) of production/ servicing is reduced (with VAI)	Measured in percentage terms- Less than 20% – 1 21–50% – 2 51–80% – 3 Above 80% – 4

(continued)

Table 4. Continued

Variables	Definitions	Operationalization
Leverage	Measure of ability to leverage VAI applications across multiple applications like internal HR processes, CSR initiatives, and many more	Scale measures extent of leverage on a scale of 1–5. 1 being low leverage and limited usage and 5 being extendible for all stakeholders
Customer experience	Holistic in nature involving the customer’s cognitive, affective, emotional, social, and physical responses to the entity, product, or service (adapted from Verhoef <i>et al.</i> , 2009)	Klaus and Maklan (2013) suggest a 7-point Likert scale with 19 items covering the dimensions of peace of mind, moments-of-truth, outcome focus and product experience, which has a reliability score of 0.93
<i>B2C adoption – use cases</i>		
Voice-commerce	Use of VAI for shopping and purchase	
Information search	Use of VAI for search related to high involvement and low involvement products, reviews and related information. This can help decrease cognitive load of searching and assimilating information	
Personal assistant	Use of VAI as a personal assistant helping to organize professional and personal life, serves as reminders, helps to pay bills and take care of mundane routine tasks. Also acts as smart home assistant to manage home appliances, security and lighting applications	
Content creation	Use of VAI for specialized content	
<i>B2C path variables</i>		
Anthropomorphism	Anthropomorphism is the phenomenon by which nonhuman entities are attributed with human-like characteristics, properties, behaviors and mental states that people believe to be uniquely human (Epley <i>et al.</i> , 2007)	Degree to which VAI application is human-like with respect to its ability to understand customers, connect with them, express and read emotions, and deal with ambiguity. Measured on 2-point scale as low and high
Net perceived benefit	Woodruff (1997), defines customer value as “a customer’s perceived preference for and evaluation of those product attributes, attribute performance, and consequences arising from use that facilitate (or block) achieving the customer’s goal and purposes in use situations	Measured as net of perceived value and perceived costs. It measures the value consumers see in VAI after accounting for privacy and security risks, individual risks and the objective costs

Source(s): Authors’ own work

licensing existing applications rather than developing new ones. When firm offerings integrate well with VAI, performance and effort expectancy are high; hence, firms easily integrate with existing platforms. Therefore, counterintuitively, higher integration potential with voice leads to lower VAI expenditure – a finding that challenges conventional technology investment logic where ease of integration typically encourages greater investment.

In the automation sector, people need a hands-free experience while driving. They want to do all music searches, [...] and demand a voice-enabled car. These applications can be [...] standard skills. I don’t want to invest in developing an expensive application if Alexa can do an equally good job. (S3-ID8).

When firms experience success with industry-specific applications requiring minimal customization while meeting expectations, they tend to revert to standard applications that are less capital-intensive and don’t challenge employee habits. This reflects usage barriers in

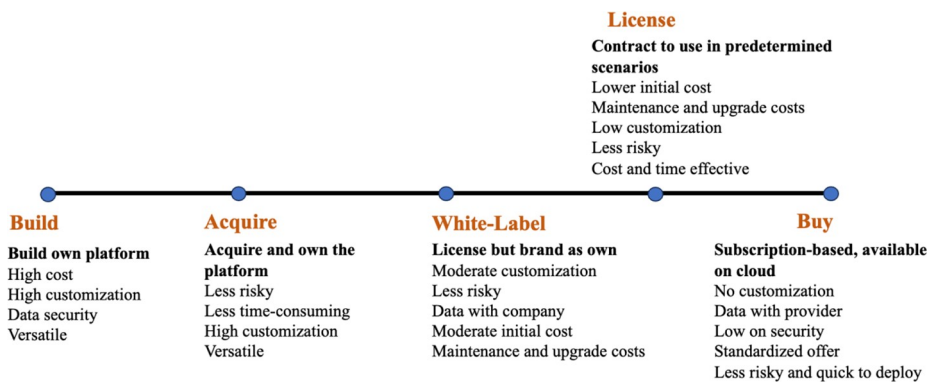


Figure 4. Possible business models based on level of investment in VAI
Source: Created by authors

Innovation Resistance Theory (Ram and Sheth, 1989). Firms prefer tested solutions as they solve existing problems, and comfort with current methods exceeds motivation to build greater capabilities. As firms embed AI in offerings, processes and decision-making, attention shifts to how software uses data that may breach privacy and security regulations (Candelon et al., 2021), leading firms to consider regulatory compliance responsibilities in VAI investments. When firms know their offerings can integrate seamlessly with AI-enabled voice technology, they invest more to safeguard and comply with regulations to build business efficiency:

We are worried about data efficiency when we use services from Amazon or Siri. We don't know if Amazon and Google will get insights at our cost [...] Hence, we wish to build our voice platform so that we know where the data rests and who uses it. (S3-ID15).

This counterintuitive relationship between integration potential and investment intensity extends the Technology Acceptance Model (Davis, 1989) by revealing how perceived ease of use can sometimes discourage rather than encourage technology investment at the organizational level. Our findings show that in organizational contexts, this relationship is moderated by competitive dynamics and standardization pressures. This aligns with institutional theory's emphasis on mimetic isomorphism (DiMaggio and Powell, 1983), where firms adopt similar practices to maintain legitimacy rather than to distinguish themselves, particularly when integration pathways are well-established:

P1. When the scope of integrating product/service solutions with VAI is high, firms are reluctant to invest more in VAI as they leverage existing solutions. However, when these firms operate in a restrictive regulatory environment, they invest more in VAI.

5.2.2 *Effect of the restrictive regulatory environment.* The Technology–Organization–Environment framework treats regulatory environments as external constraints that influence technology adoption decisions but does not adequately address how regulatory uncertainty can fundamentally override compelling business incentives (Tornatzky and Fleischer, 1990). Our research revealed three distinct ways in which regulatory environment impacts VAI investment decisions:

Criticality of the problem solved: New technologies that provide unambiguous benefits for strategic and operational effectiveness are more likely to be accepted by firms (Greenhalgh *et al.*, 2004). Firms adopt new technology to improve existing processes or solve business problems. Our research found that when issues are mundane without significant implications for firm performance, organizations resist change. Performance indicator improvement correlates with extensive use and technology's ability to address user needs, ensuring task-technology fit (Goodhue and Thompson, 1995).

Regulatory uncertainty and regulations significantly constrain technology adoption – a critical factor often underappreciated in the technology implementation literature. In restrictive regulatory environments (e.g. banking), firms must invest heavily to build proprietary VAI applications for greater data control and customize applications to fulfill data collection mandates. However, regulatory uncertainty and restrictions dampen firms' motivation for such investments, especially when adequate alternative solutions exist. Counterintuitively, firms reduce uncertainties by working with existing solutions and technology, even when VAI could solve critical problems. According to institutional theory, firms avoid adopting technology until regulatory uncertainty decreases (Frederiks *et al.*, 2022). Complex regulatory webs impose prohibitive costs on new entrants in transformative technology markets.

We are in a very highly regulated market. IRDA doesn't allow [...] So that doesn't allow us to do many AI voice technology skills. (S3-ID4).

Scope for Competitive Advantage: Firms invest in new technologies like VAI due to customer demand, interaction fluency, or new functionalities that offer sustainable competitive advantage. Ransbotham *et al.* (2017) found over 85% of executives believed AI technology would enable increased competitive advantage. When firms meet parity, investments remain low as they choose basic, less expensive applications through licensing or white labeling. However, firms seeking competitive advantage through VAI must invest heavily in acquiring or building customized VAI applications for differentiation. Firms possess heterogeneous technology capabilities with different potentials for leveraging technology competitively (Barney, 1991, 1997).

Voice technology can give a differentiated customer experience. This differentiation is needed for profitability. But it involves extensive capital investment as training for people and machines is needed. (S3-ID10).

In many cases, price and entry regulations beneficially slow technology adoption by making preemption strategies less attractive (Riordan, 1992). In restrictive or uncertain regulatory environment, only large multinational firms attempt heavy technology investments.

The partnership is needed when you are customizing a voice application or licensing such that we own the data and it only rests on our platform. [...] So, the data must be with you, [...] servers, so that it can be used with strict regulations around data privacy in the health sector. (S3-ID7).

Potential for Monetization: Refers to firms' ability to generate revenue by deploying AI-enabled voice technology across applications. This arises from:

- greater customer acquisition and retention;
- licensing or subscription offerings to other firms by investing heavily in acquiring or building voice platforms; and
- decreased costs through operational or process efficiency.

When consumers interact with technological devices, they disclose information (Puntoni *et al.*, 2021) that can be directly monetized (Lambrecht *et al.*, 2014). Predicting costs and benefits of AI applications determine the effect on firm market value. Although initial AI deployment costs are high, postimplementation benefits outweigh these costs and improve customer lifetime value (Moradi and Dass, 2022). Therefore, firms seeing greater potential for monetizing VAI applications, directly or indirectly, and confident in scaling up tend to invest more in VAI deployment.

Higher privacy protections are necessary for information disclosed in oral technology interactions (Melzner *et al.*, 2023). Firms in restrictive regulatory landscapes face higher risks in uncertain environments. They cannot fully deploy or license applications to other firms or utilize them to gain customer insights or predict behavior.

There are venture capitalists who are pumping the money into voice technology. There is nothing like- Hey! Let's invest and grow [...] a lot of regulations and risks. There is very little money left out for brands to innovate, and I don't want to spend money building something tied to many rules and regulations (S3-ID 4).

Our framework reveals that regulatory constraints operate as systematic override mechanisms that can reverse logical technology investment decisions. Hence, though individual antecedents positively influence firm investments in VAI, based on research evidence, we propose:

- P2. Firms operating in industries governed by restrictive regulations will be reluctant to invest in VAI even when such technologies can (a) help the firm solve a critical problem, (b) offer a competitive advantage, or (c) have a high potential for monetization.

5.2.3 Effect of domain-specific applications on the relationship between the level of voice AI investments and value created for business. Resource-based view theory (Barney, 1991) emphasize versatility and broad applicability as source of competitive advantage. These frameworks cannot explain when and why specialized technology might outperform generalized value creation, particularly in AI contexts. Firms realize that greater VAI investments enable customized domain specific applications that perform more efficiently in sectors like automotive and retail. AI in domain-specific applications is trained to better understand customer needs, generate insights, and suggest options superior to generalized applications. These applications perform efficiently in focused sectors, creating a self-reinforcing cycle of specialization and performance enhancement.

It's naturally a self-fulfilling loop as you build e-commerce solutions, they will be better suited that. We understand e-commerce [...] than Google. If you ask something like showing Nike shoes [...], our voice tech solution is better than Google. But if you say something like play a song, [...] . worse job than Google does. We will develop an AI-enabled [...], and use sentiment analysis to motivate them to buy or add to the cart. (S3-ID11).

While general AI implementation approaches often prioritize versatility, our research reveals that domain specialization significantly enhances perceived value by a more precise alignment between technological capabilities and specific business requirements. This specialized alignment creates a form of asset specificity (Williamson, 1985) that increases value and raises switching costs for potential users, enhancing the sustainable competitive advantage.

- P3. Firms will perceive a higher potential for business value creation when VAI includes domain-specific attributes.

5.2.4 Value creation for business. Customer experience and satisfaction emerge as central drivers of customer retention in current literature (Jain *et al.*, 2022). Research has examined how better customer experience and satisfaction can promote profitable customer retention by understanding customer lifetime value (Kumar, 2024; Alabed *et al.*, 2023). However, they do not adequately explain how VAI can create value for both firm and customer through simultaneous mechanisms. Our framework extends existing theory by demonstrating how VAI enables simultaneous value creation across multiple dimensions, with each mechanism reinforcing the other. VAI investment represents a strategic decision that enhances customer experience, delivers products and services seamlessly, and reduces operational costs. Our research findings show consequences of VAI investment reflected in: (a) better customer experience enabling customer retention and decreased acquisition costs; (b) decreased operational costs by replacing manpower for routine and automated tasks; and (c) leveraging VAI for other applications. VAI improves efficiencies and processes, enhances customer experiences, and ultimately increases revenues and perceived global brand identity.

When it comes to user and user experience, I have to wait a long time to resolve my query because the live agent is busy solving other queries. Those could be automated by the Chatbot. (S3-ID9).

Firms seek to use these applications across organizations to manage customer-facing processes like CRM, sales funnel management and complaint redressal. These enhance customer experience and increase acquisition and retention. When firms look beyond basic investment and devote resources to fully utilize VAI potential, they benefit from monetary and nonmonetary deployment.

Flipkart or BigBasket would want to build their own voice platform as they have search potential [...] platform selling stuff. The concern [...] Alexa can help firms engage and increase CLV (customer lifetime value) if people start adopting and referring because of a better experience. (S3-ID1).

P4. Investments in VAI help firms create value for the business with (a) better customer experience, (b) greater cost efficiency, or (c) the ability to leverage VAI across external and internal applications.

5.2.5 Perceived net benefit in voice AI use-cases (B2C marketplace). Customer satisfaction occurs when performance matches expectations. Satisfaction represents a relative term between individual expectations and actual situations (Franken and van Raaij, 1981). In technology studies, user satisfaction refers to feelings and attitudes derived from perceived benefits of technology interaction (Ives *et al.*, 1983). Delone and Mclean (2003) combined individual and organizational impact into “net benefits,” referring to the difference between positive and negative outcomes of technology use. Perceived net benefit represents the most appropriate measure of IS effectiveness (Ives *et al.*, 1983; DeLone and McLean, 1992) and predicts technology use (Davis, 1989; Segars and Grover, 1993). Han *et al.* (2018) proposed a positive relationship between perceived benefits and continuance intention.

Technology acceptance research treats technology as having uniform characteristics across contexts (Rogers, 2003 and Davis, 1989). However, this assumption cannot explain why same technology, especially in VAI contexts, might produce different benefit-risk evaluations across use cases. We found consumers identify four broad use cases for AI-enabled voice technology in the B2C marketplace:

- (1) *Voice commerce applications:* Ordering products hands-free, anytime, without delivery costs provide convenience. However, skepticism persists due to lack of visual confirmation.

- (2) *Content subscription*: Consumers use voice assistants to subscribe to preferred content, saving time and accessing in-depth content in chosen domains.
- (3) *Personal assistant*: VAI can be incorporated into home solutions, professional tasks and personal life management. Consumers seek empathetic applications but raise privacy concerns when applications seem intimidating.
- (4) *Search*: [Stigler \(1961\)](#) states consumers search for prepurchase information until perceived marginal benefit equals perceived marginal cost. Voice applications capitalize on understanding consumers, their purchase behavior and preferences.

Task-technology fit theory ([Goodhue and Thompson, 1995](#)) establishes that technology characteristics must align with specific task requirements to enhance user evaluation. This alignment involves sophisticated situational risk evaluation and task-specific value attribution varying systematically across use contexts ([Blut et al., 2021](#)). Consumers determine whether to continue using applications or pay based on perceived benefits and risks. Resulting satisfaction drives referrals or positive word-of-mouth.

Our research reveals a critical insight often overlooked in VAI implementation: consumer engagement ([Kumar and Pansari, 2016](#)) varies dramatically across use cases, challenging the one-size-fits-all approach in technology literature. To our knowledge, no research has investigated whether all use cases have the same perceived net benefit and customer engagement. We found that a clear continuum exists; in some applications, consumers remain unaware of nonmonetary usage costs. Consumers are more familiar with search and content creation but still exploring personal assistant use cases. They appear comfortable with relationships where they maintain control but hesitate with applications where VAIs assume control. Once fully exposed, they may find personal assistance applications intimidating and privacy-threatening, though convenience and benefits may outweigh privacy concerns. Consumers perceive greater risks than convenience benefits in voice commerce.

Our findings extend Innovation Diffusion Theory ([Rogers, 2003](#)) by demonstrating that adoption factors vary systematically across use contexts in VAI. This contextual variation challenges the common practice of treating VAI as a singular technology with uniform adoption drivers. Consumers engage in use case-specific evaluations of benefit-risk ratios, with perceived privacy risks and control factors weighing differently across contexts. This advances theoretical understanding by showing how contextual factors moderate the relationship between perceived benefits and technology engagement:

- P5. Consumer engagement with VAI will vary systematically across use cases, with: (5a) higher engagement in content subscription and search applications due to favorable perceived benefit-risk ratios; (5b) lower engagement in voice-commerce applications due to unfavorable perceived benefit-risk ratios; (5c) an inverted U-shaped engagement pattern for personal assistant applications as privacy concerns become more salient over time.

5.2.6 Effect of anthropomorphism on the relationship between perceived net benefit and consumer engagement. Anthropomorphism research suggests human engagements with technology parallel human-human engagement patterns (e.g. [Reeves and Nass, 1996](#); [Nass et al., 1999](#)). [Dellaert et al. \(2020\)](#) proposed that consumer psychological comfort with VAI depends on the alignment between decision task type and human likeness ([So et al., 2024](#)).

Our research dramatically advances the understanding of anthropomorphism effects by revealing highly nuanced and context-dependent impacts that significantly depart from

existing literature that treats anthropomorphism as uniformly beneficial. Recent research demonstrates differential anthropomorphism effects across contexts. Consumers prefer human-like attributes for applications involving personal or social identity but less anthropomorphic technology for task-oriented, transactional and objective applications (Blut *et al.*, 2021; Moriuchi, 2019; Ferrari *et al.*, 2016; Luo *et al.*, 2019; Zarouali *et al.*, 2018) but fails to provide a nuanced understanding of the how varying levels of anthropomorphism impact customers' perceived benefit across various contexts. Our research revealed varying views on VAI anthropomorphism. While it enhances some applications, it inhibits others.

Alexa is warm, and it is important how it handles non-preconfigured Q&A. Each VA's personality will be different, determining how people will use and relate to it. The human likeliness [...] determine how people will use it and for what. (S3-ID2).

In VAI applications, consumers fear losing privacy and data security in ambient environments. Schweitzer *et al.* (2019) proposed that customers describe their VAI relationship as servant–master. Increased interactions occur when consumers feel superior to devices. Extended use of VAIs anthropomorphized as partners negatively impacted future usage intentions.

Personalization and emotions are important. People using VAs feel more comfortable because their personal information is with technology and not a person! So, they want nonintrusive technology. If it becomes humanlike, it will make consumers uncomfortable (S3-ID4).

P6. The effect of anthropomorphism on consumer engagement will be dependent on use case, such that higher anthropomorphism will: (6a) positively affect engagement in content subscription and search contexts; (6b) negatively affect engagement in voice commerce contexts; (6c) show a curvilinear relationship in personal assistant applications.

Our findings collectively reshape understanding of VAI implementation and value creation in fundamental ways. The framework reveals that VAI adoption is governed by counterintuitive relationships between integration potential and investment levels, subject to powerful regulatory constraints that override business incentives, enhanced through domain specialization rather than generalization, and moderated by context-dependent consumer responses to anthropomorphism. These insights demand a more nuanced, contextual approach to VAI strategy that considers the specific regulatory environment, use case characteristics, and appropriation mechanisms rather than applying uniform implementation frameworks.

6. Conclusions and implications

AI-enabled voice technology offers significant potential to create value in both B2C and B2B marketplaces, yet firms struggle to monetize their VAI investments. Our study examines how organizations can create and appropriate value using VAI by adopting suitable business models that build global identity (Iglesias *et al.*, 2020). Using grounded theory, we collected data from three studies across customers, employees, organizations and decision-makers to propose a Voice AI value creation and appropriation framework.

Our research reveals that VAI implementation requires fundamentally different strategic approaches compared to traditional technology deployment frameworks. Table 5 summarizes how our findings challenge conventional wisdom across seven critical dimensions of technology implementation. By contrasting traditional approaches with our

Table 5. Reimagining technology implementation: traditional approaches vs. VAI framework insights

Strategic dimension	Traditional technology implementation approaches	Our VAI framework insights	Managerial implications
Integration and investment	Higher integration potential leads to greater investment as firms capitalize on seamless adoption opportunities	Paradoxical relationship: Higher integration potential leads to <i>lower</i> investment in proprietary solutions as firms leverage existing platforms	Begin with generic, widely-used solutions to penetrate markets before advancing to domain-specific solutions; expect lower price premiums for easily integrated applications due to standardization pressures
Regulatory impact	Regulatory considerations are treated as secondary factors in technology adoption decisions	Regulatory constraints create powerful counterforces that override otherwise compelling business incentives	Conduct thorough regulatory impact assessments before VAI investments; prioritize data governance frameworks; prepare for asymmetric implementation timelines across industries
Domain specification	Technology implementations prioritize versatility and broad applicability	Domain specialization significantly enhances perceived value and competitive advantage	Focus VAI development on industry-specific use cases rather than general applications; build domain expertise into AI training processes; expect higher monetization potential from specialized solutions
Value creation mechanisms	Value creation is conceptualized primarily as cost reduction or process enhancement	Value creation emerges through multiple pathways: enhanced customer experience, operational efficiency, and cross-application leverage	Design VAI initiatives with multiple value creation objectives; implement metrics that capture both tangible and intangible benefits; establish cross-functional teams to maximize value capture
B2B and B2C integration	B2B and B2C technologies are developed and implemented as separate initiatives	B2B and B2C marketplaces create bidirectional influences that shape implementation success	Integrate consumer insights into B2B application development; leverage B2B implementation learnings to enhance consumer experiences; coordinate implementation strategies across both domains
User experience design	Uniform user experience principles applied across implementation contexts	Context-dependent anthropomorphism effects require calibrated design approaches	Tailor anthropomorphic characteristics to specific use cases; increase human-like features in content and search applications; reduce them in commerce and personal assistant contexts
Monetization strategy	Direct revenue generation treated as primary success measure	Multiple monetization pathways including indirect value through customer acquisition and retention	Develop comprehensive value appropriation frameworks that capture both direct and indirect benefits; align monetization mechanisms with use case characteristics

Source(s): Authors' own work

6.1 Implications for theory

The appeal of VAI is evident, whether as an opportunity to reduce costs and enhance customer engagement efficacy from the firm perspective or to decrease transaction costs and enhance brand engagement from the customer perspective. However, the optimal pathway to value creation using VAI remains unclear. The paper makes significant contribution to existing work, highlighting the evolving nature of VAI.

Established value creation frameworks, including Technology Adoption Model (Davis, 1989), Resource-Based View (Barney, 1991) and Value co-creation model (Grönroos, 2011), typically assume positive relationships between technological capabilities and value appropriation potential. However, evidence from VAI implementations reveals that when firms can easily integrate VAI solutions, they paradoxically reduce investment intensity, preferring standardized solutions over customized development. This counterintuitive dynamics suggests that conventional value creation logic – where ease of implementation encourages investment – does not apply to VAI contexts. Instead, competitive standardization pressures and the availability of white-label solutions appear to depress innovation incentives precisely when integration potential is highest.

Rogers' (2003) Innovation Diffusion Theory treats technologies as having relatively uniform characteristics that influence adoption patterns consistently across contexts. Our research indicates that anthropomorphism produces dramatically different effects across use contexts: enhancing engagement in information search applications while reducing comfort in transactional scenarios. This context-dependent variability suggests that VAI is a technology platform whose perceived attributes and adoption drivers vary systematically across applications.

Our findings suggest that regulatory environments create unique dynamics in VAI adoption that override traditional business incentives. Unlike conventional technologies where regulatory compliance represents an implementation consideration (The TOE framework – Tornatzky and Fleischer, 1990), VAI's data collection and processing capabilities make regulatory constraints fundamental determinants of investment approach – often compelling firms to choose expensive proprietary development despite compelling business cases for standardized solutions.

The framework highlights the importance of domain-specific applications for creating value using VAI. Most existing research focuses on consumer adoption of VAI and outcomes like engagement and continued use. Limited work addresses firm behavior toward VAI and value creation through AI-enabled voice technology (Kohtamäki and Rajala, 2016). Our research fills this critical gap by extending AI technology adoption research to postadoption value creation and appropriation processes.

Our framework proposes that value creation and monetization in VAI require the integration of B2B and B2C marketplaces, while creating value differently in these marketplaces. While monetization potential and value creation scope are higher in B2B applications, consumer pull from the B2C marketplace critically influences VAI investment decisions. To our knowledge, no previous research has examined the interaction between these marketplaces for VAI monetization. Table 6 gives a summary of how differential value is created in B2B versus B2C contexts.

This research bridges IS and marketing literature in the VAI context. Unlike other technologies typically pushed by firms to customers, VAI gained consumer acceptance first, giving businesses confidence to adopt it. Our contribution to AI technology monetization

Table 6. Differential value creation mechanisms: B2B versus B2C contexts

Dimension	B2B context	B2C context	Interdependencies
Primary value creation	• Licensing VAI platforms to other firms • White-label solution development • Domain-specific application customization • Operational efficiency gains	• Enhanced customer engagement • Reduced transaction friction • Personalized content delivery • Hands-free convenience benefits	B2C adoption comfort creates confidence for B2B investment decisions
Monetization mechanisms	• Subscription-based platform access • Per-transaction licensing fees • Custom development contracts • Data analytics services	• Premium feature subscriptions • Voice commerce transactions • Advertising revenue through voice channels • Enhanced customer lifetime value	Consumer willingness to pay validates B2B value propositions
Investment drivers	• Competitive advantage through differentiation • Critical problem-solving capabilities • Regulatory compliance requirements • Cost reduction potential	• Perceived net benefits across use cases • Anthropomorphic appeal (context-dependent) • Privacy-convenience trade-offs • Use case-specific adoption patterns	High consumer engagement signals market readiness for B2B solutions
Implementation approach	• Build vs. buy decisions based on integration complexity • Domain specialization over generalization • Ecosystem-driven development	• Context-dependent anthropomorphism preferences • Privacy control expectations • Seamless integration with daily routines	Consumer adoption patterns inform B2B implementation strategies
Value appropriation barriers	• Regulatory uncertainty dampening investment • Standardization pressure reducing differentiation • High development costs vs. white-label options • Data privacy compliance requirements	• Privacy concerns limiting engagement • Inconsistent performance across use cases • Anthropomorphism creating discomfort in certain contexts • Limited visual confirmation capabilities	Consumer resistance creates B2B adoption hesitancy

(continued)

Table 6. Continued

Dimension	B2B context	B2C context	Interdependencies
Success metrics	• Revenue generation from licensing/subscriptions • Cost savings from operational efficiency • Market share in domain-specific applications • Customer acquisition through VAI capabilities	• Engagement frequency and duration • Task completion success rates • Customer satisfaction scores • Retention and referral patterns	B2C engagement metrics validate B2B value creation potential
Strategic positioning	• Platform provider vs. solution integrator • Proprietary development vs. ecosystem participation • Industry specialization vs. broad application • Partnership vs. competition strategies	• Convenience provider vs. relationship builder • Privacy-respecting vs. data-leveraging approaches • Task-focused vs. conversational interaction styles • Multi-platform vs. single-ecosystem adoption	Consumer platform preferences influence B2B ecosystem strategies

Source(s): Authors' own work

literature emphasizes the necessity of value co-creation among multiple stakeholders while illustrating pathways to monetization.

Additionally, our research contributes new insights to consumer-focused voice technology literature, proposing that consumers' perceived net benefits vary across use cases and are differentially affected by VAI anthropomorphism. This relationship warrants further investigation, adding a novel dimension to existing literature.

6.2 Implications for practice

Our research offers several actionable insights for organizations implementing VAI across three key domains: strategic positioning, implementation approach and consumer experience design.

6.2.1 Strategic positioning. VAI technology offers distinct strategic advantages over conventional voice technology: increased accuracy across languages, reliable voice bots and virtual assistants, and predictive speech assistance. These capabilities enable businesses, particularly in developing economies, to reach underserved populations including less educated demographics, rural areas and consumers with limited reading abilities or technology interaction challenges.

In B2B voice applications, customer experience enhancement and cost efficiency emerge as primary expected outcomes. Delivering superior customer experiences through VAI enables brands to strengthen market positions while demonstrating technological adaptability, ultimately supporting both customer acquisition and retention objectives.

Firms with prior AI experience demonstrate greater willingness to embrace VAI when they perceive competitive advantage opportunities. Similarly, when organizations believe

VAI can solve critical business problems, previous experience with AI or voice technology accelerates deployment decisions.

6.2.2 Implementation approach. Our findings reveal a counterintuitive relationship between integration potential and investment intensity. More intuitive VAI integration opportunities correlate with reduced firm investment because widely recognized integration possibilities often have white-label or licensing solutions, reducing the need for indigenous development. This suggests a strategic sequencing approach: firms developing AI-enabled voice applications should begin with generic, widely used solutions to penetrate markets and build equity before advancing to domain-specific solutions. Generic solutions command lower price premiums due to competition but remain essential for building customer equity and differentiated global brand identity.

Organizations committed to substantial VAI investment benefit from developing domain-specific solutions. When pursuing solutions specific to their business challenges, investment willingness increases significantly. Therefore, firms seeking long-term VAI advantages should begin with domain-specific deployments to enhance customer value and monetize investments through licensing or white-labeling within their sector.

Notably, regulatory constraints have emerged as a powerful counterforce that dampens firm interest in deploying VAI despite compelling business incentives. As governments worldwide develop AI regulatory frameworks, many organizations are adopting wait-and-see approaches, temporarily decelerating innovation. This highlights the delicate balance regulators must maintain between safeguarding the future and fostering innovation during this critical period of technological acceleration.

6.2.3 Consumer experience design. For consumers, VAI introduces new dimensions of anthropomorphized user-offering connections. Our research reveals that the boundary between helpful and intrusive technologies varies significantly by use case. In some domains, consumers view VAI as a conversational partner; in others, merely as a listener. This distinction is crucial because when users perceive VAI as listening tools, focus remains on the user without attributing humanlike thought or memory to the technology. Conversely, when users view VAI as conversational tools, they sacrifice privacy control as information enters the VAI's memory.

As firms integrate increasingly sophisticated memory and thought capabilities into VAI, users become more aware of the distinction between conversational and listening interactions. Paradoxically, organizations' efforts to assist users through enhanced capabilities may make VAI seem uncomfortably helpful or intrusive. While anthropomorphism enhances satisfaction with VAI in content subscription and search domains, it may reduce satisfaction in personal assistant and e-commerce contexts due to privacy concerns diminishing perceived benefits.

These findings suggest that firms should calibrate anthropomorphic characteristics according to use case, rather than applying uniform design approaches across all VAI applications. This context-sensitive implementation strategy will maximize engagement while minimizing privacy concerns that could undermine adoption.

6.3 Limitations and future research

While our research provides valuable insights into VAI value creation and appropriation mechanisms, two primary limitations warrant acknowledgment and suggest directions for future research.

Our theoretical framework, while empirically grounded, requires formal empirical validation. The propositions we have developed represent testable relationships that should be examined through both qualitative and quantitative methodologies across diverse

organizational contexts. Particularly important is the validation of the counterintuitive relationships we identified between integration potential and investment levels, and the moderating effects of regulatory environments on VAI implementation decisions. Future research employing statistical analysis of attitudinal and behavioral data could address this limitation while further refining our understanding of these complex relationships.

The framework necessarily simplifies the heterogeneity in organizational contexts and decision-making processes that influence VAI implementation. Factors including decision-makers' technology orientation, firm size, firm risk appetite, organizational culture and performance metrics likely impact VAI investment levels in ways that warrant more detailed investigation. Future research should explore how these organizational characteristics moderate the relationships identified in our framework, potentially through comparative case studies or configurational analyses that can capture complex patterns of contextual factors.

Our proposition regarding regulatory constraints as a powerful counterforce to VAI adoption deserves focused investigation. Future studies could examine how specific regulatory frameworks (e.g. GDPR in Europe versus CCPA in California) differentially impact VAI implementation decisions and whether these effects vary across industries with different sensitivity to data privacy concerns.

The context-dependent effects of anthropomorphism warrant rigorous experimental validation. Researchers could design controlled experiments to test how varying levels of anthropomorphism influence user engagement across different use cases, potentially revealing optimal anthropomorphic calibration for specific VAI applications.

Longitudinal research examining VAI's influence on business value, customer acquisition, and retention through behavioral data analysis would significantly advance understanding of long-term value creation mechanisms. Similarly, cross-cultural studies comparing VAI adoption patterns across regions with different privacy orientations, technological infrastructures and cultural attitudes toward AI could reveal important contextual boundaries to our framework.

Finally, sector-specific investigations testing our framework across manufacturing, financial services, healthcare and hospitality would enhance its practical utility while potentially revealing industry-specific modifications. These focused explorations would refine our understanding of VAI value creation while providing increasingly tailored guidance for practitioners navigating this complex technological landscape.

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Further reading

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Supplementary material

The supplementary material for this article can be found online.

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